Project Summary Report

Travis County Congressional Districts Analysis- A study of Possible Gerrymandering

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Background

The United States Constitution set up the framework for every state to have representation in the United States House of Representatives. The number of House of Representatives has been capped at 435 members since the early 1900's (Figure 1) . House seats are reapportioned after each decennial census with some states gaining and others losing seats. Each district should closely be of equal population within the state (Congressional Apportionment, 2018). Any population difference greater than one percent between the most and least populated districts within the state will likely be unconstitutional. In this study, VTD will be used for the term "Voting Tabulation Districts".



Figure1. United States Congressional Districts

For the past 25 years, since Texas became what is considered a "red state" with all statewide offices and both the Texas House and Senate as

majority Republican, Travis county has remained a bastion of a more liberal political leanings and has remained mostly a "blue" county. For example, in the November 2016 election, Travis county voted 66.94% Democratic to 30.01% Republican. The presidential election of 2016 followed a similar pattern with 65.77% voted for Clinton to 30.01% Trump (Dana DeBeauvoir-Travis County Clerk, 2018).

Travis county is split (shared) between five congressional districts: 10th (Michael McCaul-R), 17th (Bill Flores-R), 21st (Lamar Smith-R), 31st (John Carter-R), and 35th (Lloyd Doggett-D) (Directory of Representatives, 2018). This means that Travis County is represented as 80 percent republican (not by area, but by representation). A discrepancy is seen when comparing the results of elections with representation in Travis county versus the rest of the district outside Travis County. A deeper look will tell us exactly how far off this is. Figure 2 shows how Travis County compares with the rest of the state in the 2012 and 2016 Presidential Elections. Figure 3 and Table 1 show Travis County's voting record per VTD for both the 2012 and 2016 elections as compared with the population density for the county (Texas Legislative Council, 2018). It is clear from here how democratic (blue) Travis county is politically.

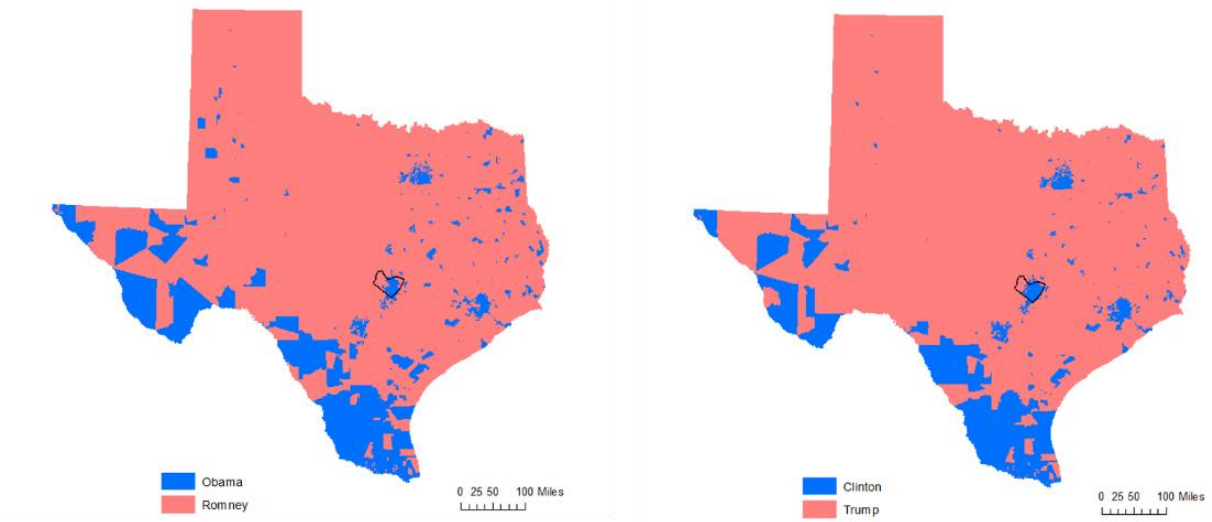


Figure 2. Comparing the 2012 and 2016 Presidential Elections Results In Texas In Relation to Travis County

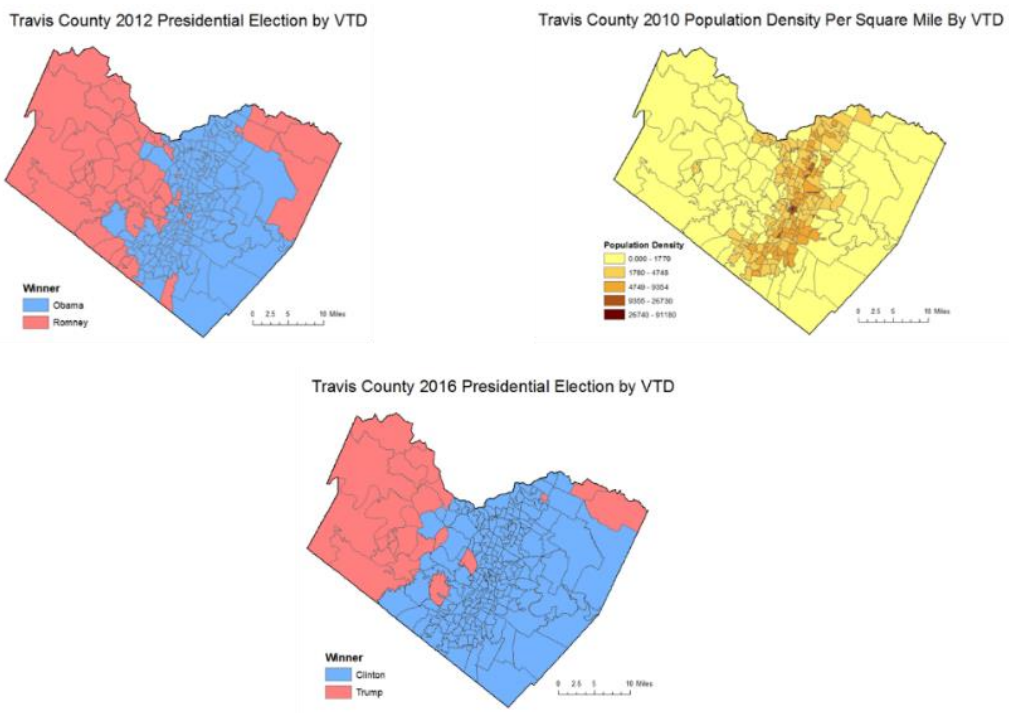


Figure 3. Travis County Partisan Breakdown for 2012 and 2016 Presidential Elections Compared with Population Density

Table 1. Travis County 2012 and 2016 Presidential Election Results

Year	Straight Ballot (%)	Presidential Election (%)
2012	Democratic: 61 Republican: 36	Obama: 60 Romney: 36
2016	Democratic: 66 Republican: 30	Clinton: 65 Trump: 30

Gerrymandering is the deliberate manipulation of representative districts for the benefit of the party designing the district. Two major tools of gerrymandering are "packing" and "cracking" (Lapowsky, 2018). In packing, the designer packs as many of the opposing party's voters into as few of voting districts as he or she can. He or she then implements cracking by spreading the remainder of these voters to as many other districts friendly to the designer's party as possible. Figure 4 illustrates how groupings can be manipulated to favor one side, even if it's the minority.

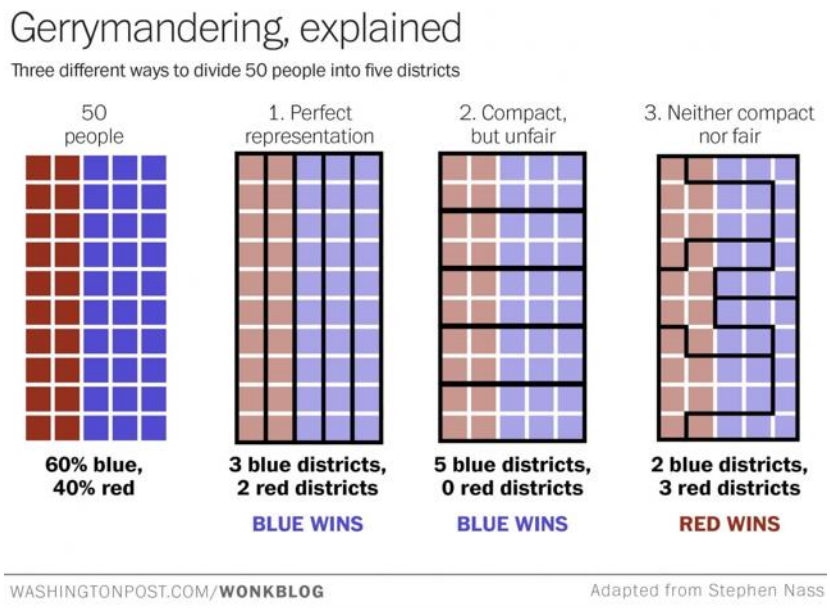


Figure 4. Simple Example of Gerrymandering

(Ingraham, 2015)

Typically, the more elongated a shape is, the less representative it is. The circular index is one quantitative way of calculating whether a particular shape of a district is considered representative. In this study, the "circular index" will be calculated for each US Congressional district representing Travis County. The formula used for this calculation is

$$circular\ index = perimeter \div (2 \times \pi \times \sqrt{(area \div \pi)})$$

$$Formula\ 1$$

For a perfect circle, the circular index is 1. For analyzing representative districts, a circular index greater than 1.67 is considered "irregular" (Morrill, 1981).

Goals

By comparing the racial breakdown and voting records between the congressional districts within Travis county versus the rest of the congressional district outside of Travis county, the study's goal is to test to see if the part within Travis county is consistent with the rest of the district or if there is a measurable difference. A measurable difference will indicate possible gerrymandering taking place.

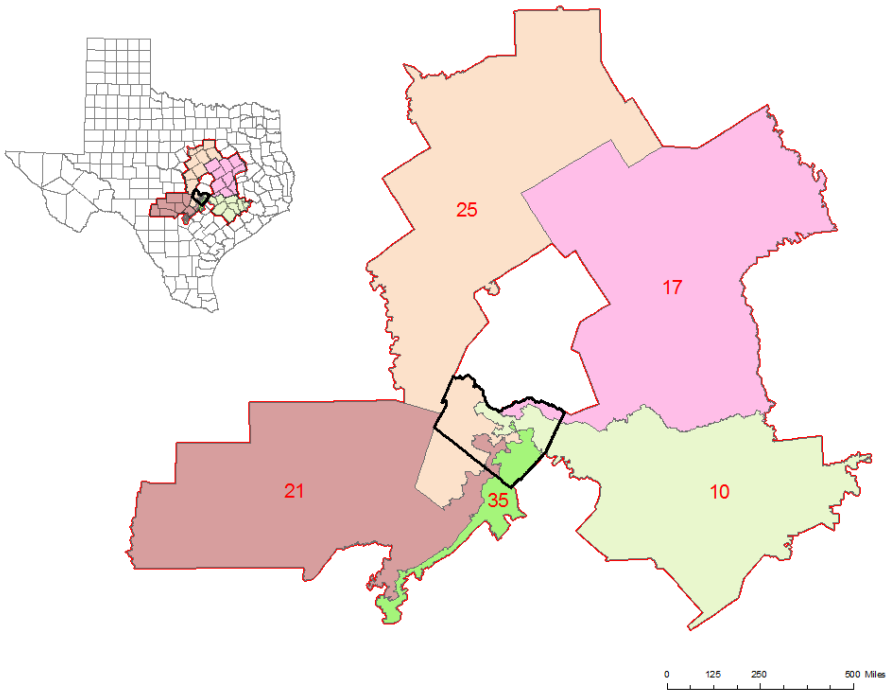


Figure 5. United States Congressional Districts Representing Travis County

The goals can be summarized in two research questions: *Is there a significant difference between the racial and political breakdown of each congressional district in question in Travis County with the portion of the district outside Travis County?* If the answer is "Yes", the data will quantify by how much it is and could suggest that gerrymandering is taking place within Travis County. The racial breakdown of the congressional districts will be found by analyzing the census blocks contained within the districts utilizing the VTD files and the voting habits will be found from the results of the 2016 presidential election for each county or voting precinct. The second research question is: *Do the shapes of the congressional districts representing Travis County violate any standards and considered gerrymandered?*

Results

Figure 6 and Figure 7 show the results of the analysis done comparing the portions of each district within Travis County versus the portions outside Travis County for the 2016 Presidential Election. Figure 6 quantitatively

shows what is seen in Figure 7 that partisan gerrymandering is highly probable in each of the five districts.

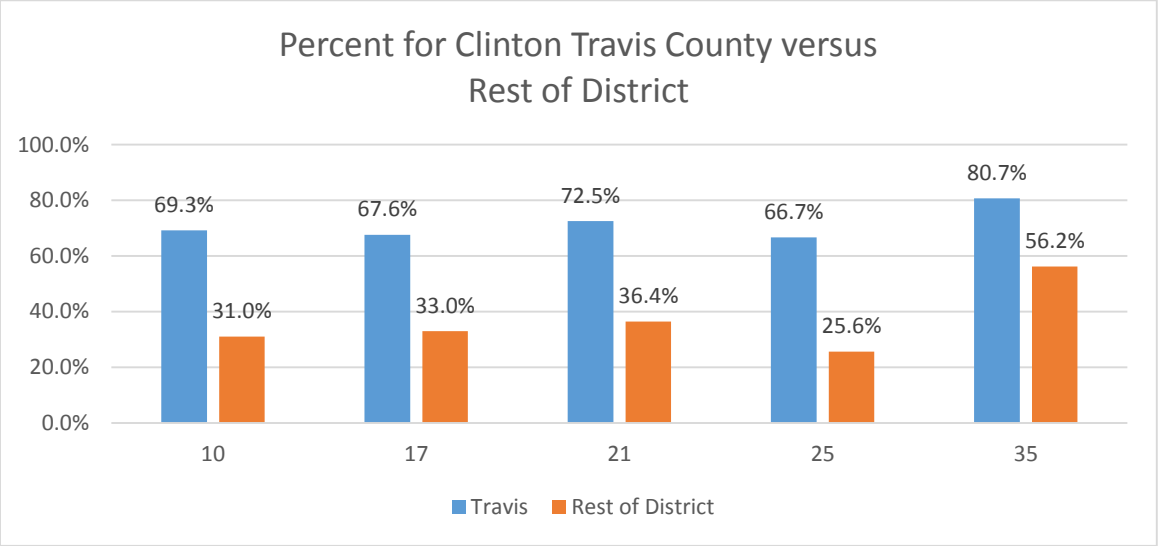


Figure 6. Comparing 2016 Presidential Election Results Within Travis County and Outside Travis County For All Congressional Districts Representing Travis County.

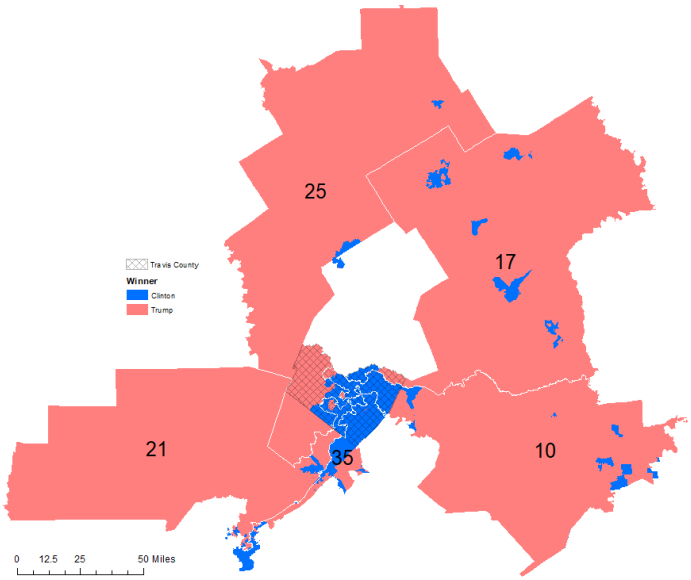


Figure 7. Map Comparing 2016 Presidential Election Results Within Travis County and Outside Travis County For All Congressional Districts Representing Travis County.

Figures 8-13 show the results of the analysis done comparing the portions of each district within Travis County versus the portions outside Travis County for white, Hispanic and black populations. The largest discrepancy is the

comparing the black population density in district 35 where the population density within Travis County is 13.9% and outside Travis County is 5.4%.

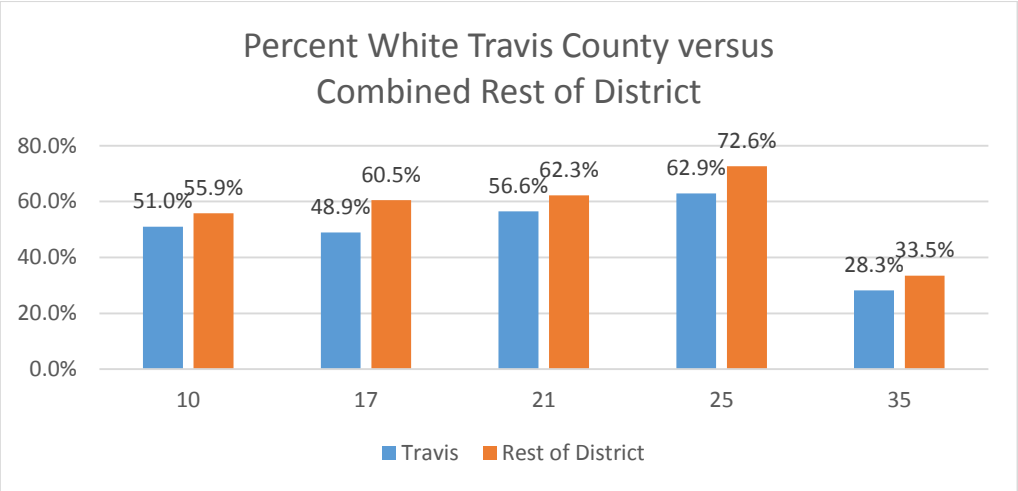


Figure 8. Comparing Percentage White Population Within Travis County and Outside Travis County For All Congressional Districts Representing Travis County.

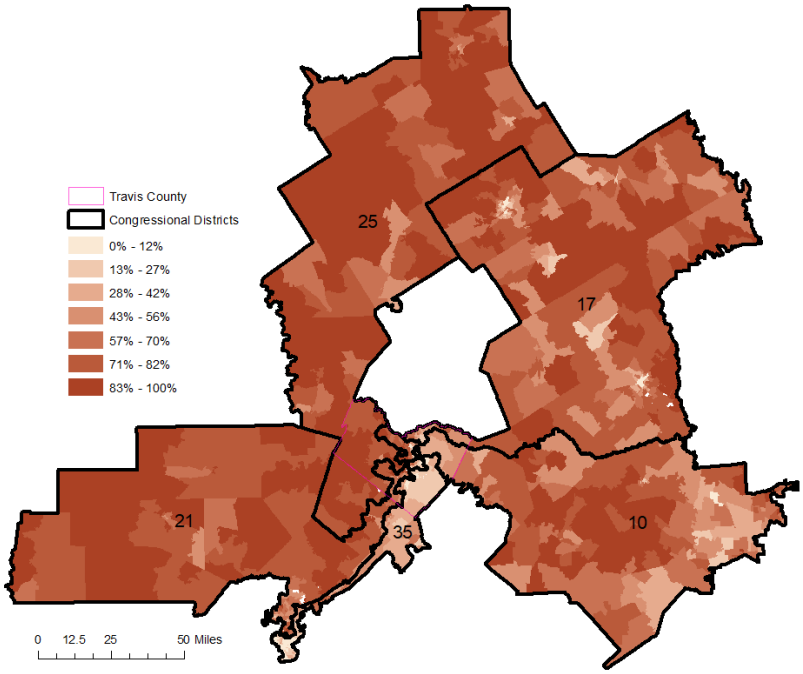


Figure 9. Map Comparing Percentage White Population Within Travis County and Outside Travis County For All Congressional Districts Representing Travis County.

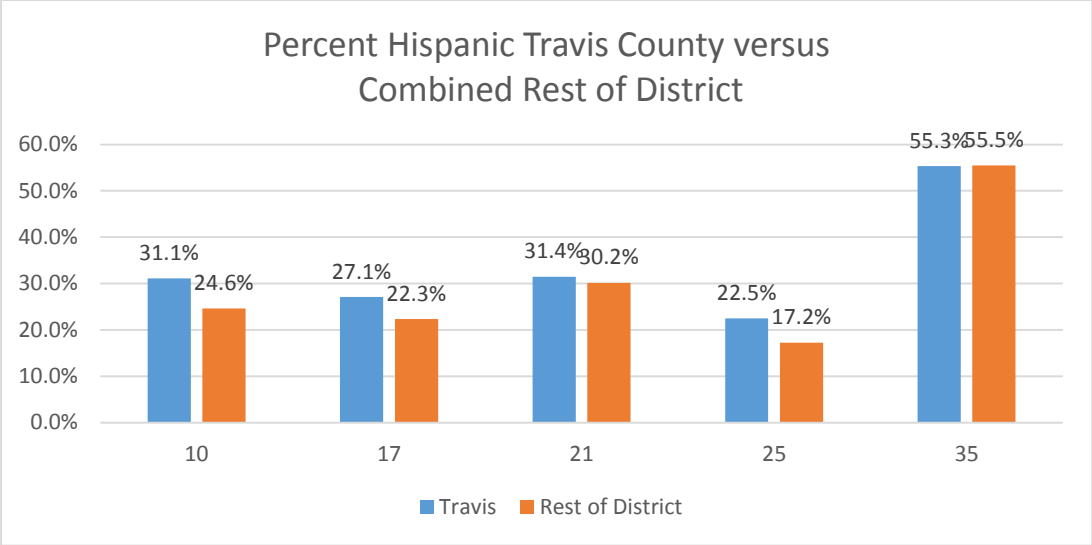


Figure 10. Comparing Percentage Hispanic Population Within Travis County and Outside Travis County For All Congressional Districts Representing Travis County.

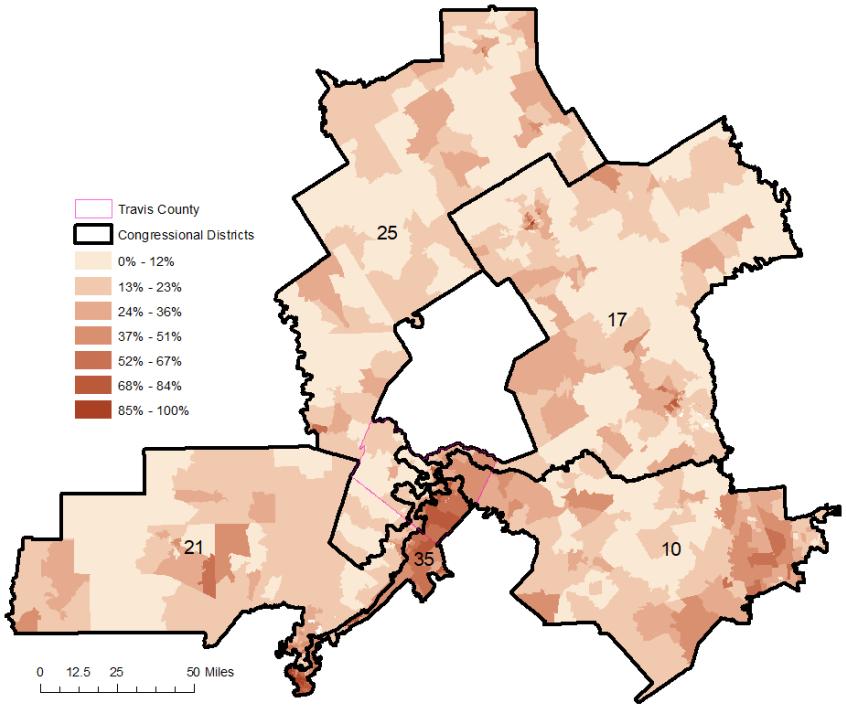


Figure 11. Map Comparing Percentage Hispanic Population Within Travis County and Outside Travis County For All Congressional Districts Representing Travis County.

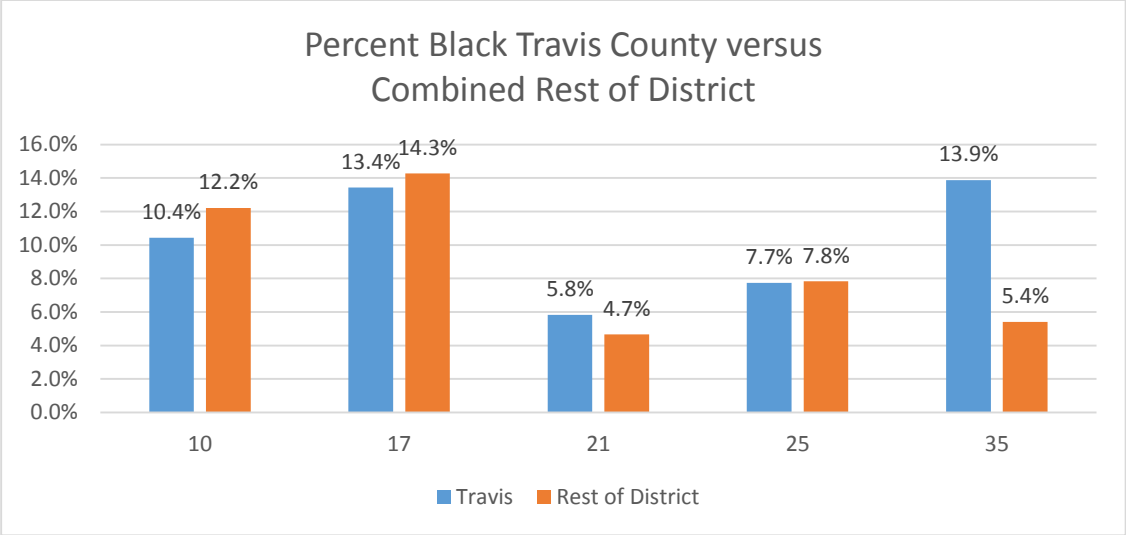


Figure 12. Comparing Percentage Black Population Within Travis County and Outside Travis County For All Congressional Districts Representing Travis County.

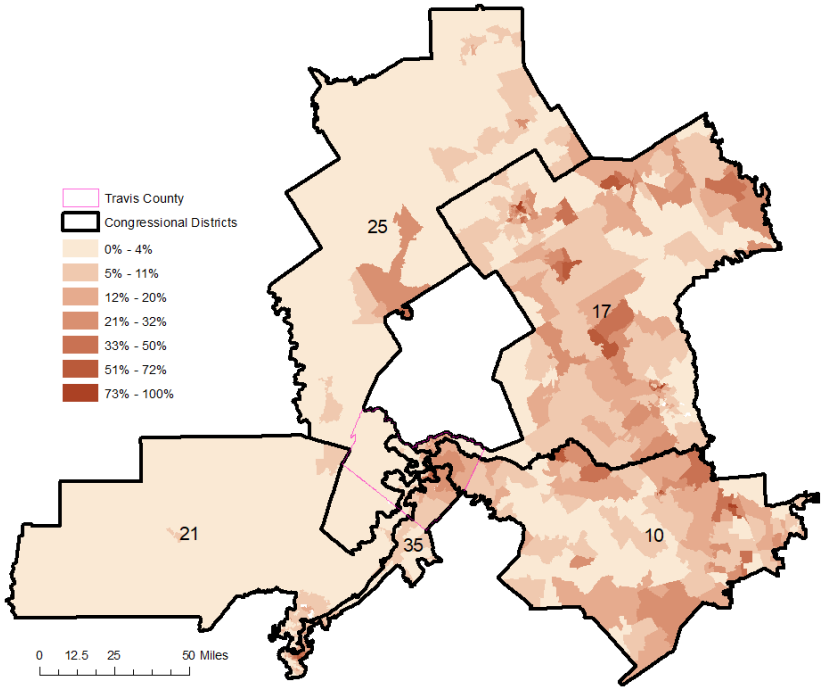


Figure 13. Map Comparing Percentage Black Population Within Travis County and Outside Travis County For All Congressional Districts Representing Travis County.

Table 2 shows the results of the circular index calculation for each Congressional District with input of the area and perimeter for each shape. All of the districts fail the representation test if 1.67 is the cutoff, but District 35 is by far the least representative shape between all of the districts.

Table 2. Circular Index for Each Congressional District in Travis County

District	Representative and Party	Area (Sq. Mi)	Perimeter (Mi)	Circularity Index
10	Michael McCaul-R	5143.7	632.3	2.49
17	Bill Flores-R	7768.8	693.3	2.21
21	Lamar Smith-R	5956.8	617.9	2.26
25	John Carter-R	7744.4	774.5	2.48
35	Lloyd Doggett-D	601.5	367.4	4.23

Figure 14 singles out District 35 as it is the one district in this study that violates both partisan and racial gerrymandering criteria along with it's oblong shape making it unrepresentative. This district is currently under scrutiny and is currently in the court system.

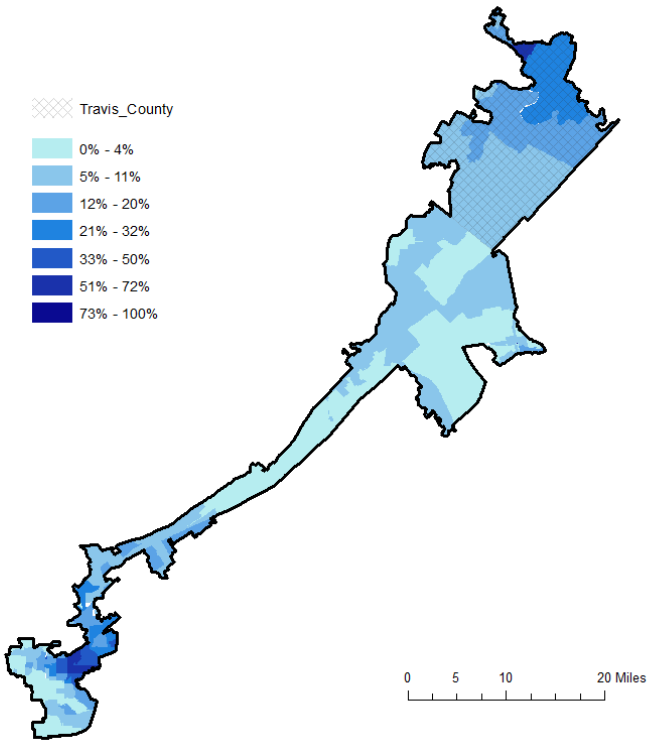


Figure 14. House District 35 Percentage Black-Expanded.

Conclusions

It is evident from the data collected and summarized in Table 3 that there is indeed partisan gerrymandering taking place with all of the districts and at least one case of racial gerrymandering with district 35. Also the shapes of the districts are drawn in a manner that is not representative

Table 3. Summary of Findings in Study

District	**Party**	**Circular Index**	**2016 Presidential Election Percentage Clinton**		**Racial Representation Percentage**					
			Travis	*Dist*	*Travis*	*Dist*	*Travis*	*Dist*	*Travis*	*Dist*
					White	*White*	*Hisp*	*Hisp*	*Black*	*Black*
10th	R	2.49	69.3	31.0	51.0	55.9	31.1	24.6	10.4	12.2
17th	R	2.21	67.6	33.0	48.9	60.5	27.1	22.3	13.4	14.3
21st	R	2.26	72.5	36.4	56.6	62.3	31.4	30.2	5.8	4.7
31st	R	2.48	66.7	25.6	62.9	72.6	22.5	17.2	7.7	7.8
35th	D	4.23	80.7	56.2	28.3	33.5	55.3	55.5	13.9	5.4

Research Questions / Answers:

Is there a significant difference between the racial and political breakdown of each congressional district in question in Travis County with the portion of the district outside Travis County?

- There is indeed a significant difference in how voters within Travis county vote as compared to the rest of the Congressional District.
- No significant differences between the racial breakdown between the population living within Travis County as compared with the rest of the

Congressional District with the exception of District 35, where the percentage of the black population inside Travis County is over twice of that outside Travis County.

Do the shapes of the congressional districts representing Travis County violate any standards and considered gerrymandered?

- If we go by 1.67 as the maximum value allowed for a Congressional District, then all of the Congressional Districts violate the acceptable roundness criteria.
- Congressional District 35 is especially misshaped for representation.
- It is no surprise that Congressional District 35 is under scrutiny and we are currently waiting a court ruling to see whether it needs to be redrawn or not.

Recommendations

It is highly recommended that the shapes of the United States congressional districts representing Travis County be redrawn in such a way to more accurately represent the county. The data shows that Travis County lacks representation based on packing and cracking indicative of gerrymandering. With so many court cases currently in place, it is hopeful that something will be done to address this and other examples of gerrymandering.

Acknowledgments

I would like to thank Sally Holl from ACC with her guidance in the entire project and with her help in setting up the personal database which was key to the analysis. I would also want to thank Charles Eckstein from the Texas Legislative Council with his help in guiding me to the VTD shapefiles and the 2010 census population data combined with election results.

Tools and Settings Used in Study

- ArcMap 10.5.1
- File Geodatabase: Travis_County_Cong.gdb
- Personal Geodatabase: Gerrymandering.mdb

- Coordinate System: NAD1983 StatePlane Texas Central FIPS 4203 Feet
- Projection: Lambert Conformal Conic

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